

Digital Collateral in Decentralised Finance: Credit, Control, and Conflict of Laws¹

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Abstract

This paper examines how Bitcoin-collateralised lending restructures the architecture of credit, collateral, and enforcement within decentralised finance. It argues that DeFi lending protocols do not merely introduce new financial instruments, but relocate functions traditionally performed by legal and financial intermediaries—including collateral enforcement, liquidity discipline, and risk allocation—into automated systems of protocol governance.

Drawing on digital asset regulation and private international law, the article analyses how over-collateralisation, algorithmic liquidation, smart-contract custody, and continuous protocol monitoring generate forms of risk discipline distinct from traditional banking credit. In these environments, solvency depends less on borrower creditworthiness than on verifiable collateral and automated execution.

The work further argues that this transformation exposes structural limitations within existing legal frameworks. Regulation (EU) 2023/1114 on Markets in Crypto-Assets (MiCA) establishes obligations for centralised crypto-asset service providers but leaves fully decentralised collateral structures largely outside its regulatory perimeter. More fundamentally, decentralised lending destabilises the territorial assumptions underlying private international law: digital collateral lacks a stable situs, protocol governance operates across distributed infrastructures, and smart-contract enforcement challenges conventional jurisdictional and insolvency frameworks.

By situating Bitcoin-backed lending within the broader transformation of transnational financial governance, the article contends that decentralised collateral systems require not only regulatory adaptation, but the development of a coherent conflicts architecture capable of addressing the proprietary, jurisdictional, and enforcement implications of borderless digital assets.

Keywords: Decentralised Finance (DeFi), digital collateral, conflict of laws, technological control, algorithmic liquidation, MiCA Regulation, Private International law.

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I. INTRODUCTION

Decentralised finance (DeFi) is frequently described as a technological innovation in financial markets. Yet its most significant transformation may be institutional rather than technological. Bitcoin-collateralised lending does not merely create new forms of credit; it restructures the architecture of collateral, enforcement, and risk allocation by relocating functions traditionally performed by financial intermediaries and legal institutions into protocol-based systems of governance.²

In conventional credit markets, lending relationships depend upon institutional trust, intermediary supervision, and territorially grounded systems of legal enforcement. Borrower solvency is assessed through discretionary underwriting, while collateral rights ultimately rely upon judicial enforcement and institution-based risk management.³ DeFi lending protocols alter these assumptions fundamentally. In decentralised collateral systems, risk management is embedded in code, collateral enforcement is automated through smart contracts, and solvency is maintained through predefined collateral mechanisms rather than institutional supervision.⁴

Bitcoin-collateralised lending provides one of the clearest illustrations of this structural transformation. Borrowers obtain liquidity without liquidating their digital assets, while lenders rely not on reputational guarantees or judicial enforcement, but on over-collateralisation, automated liquidation, and transparent protocol rules.⁵ The resulting system produces a form of endogenous risk discipline in which leverage is constrained not by institutional discretion, but by verifiable collateral and automated execution.

This transformation carries significant legal implications. Existing regulatory and Private International law frameworks were designed for financial systems structured around identifiable intermediaries, territorial localisation, and legally attributable acts. Decentralised collateral systems destabilise all three assumptions simultaneously.⁶ Smart-contract governance operates across distributed infrastructures; crypto-assets lack a stable situs; and automated liquidation mechanisms challenge traditional distinctions between contractual performance, proprietary enforcement, and insolvency administration.

Regulation (EU) 2023/1114 on Markets in Crypto-Assets (MiCA) represents the most comprehensive attempt to regulate digital asset markets within the European Union. However, MiCA largely governs the perimeter of decentralised finance—particularly custodial intermediaries and crypto-asset service providers—while leaving fully decentralised collateral

²Fabian Schär, *Decentralised Finance: On Blockchain- and Smart-Contract-Based Financial Markets*, 103 Fed. Reserve Bank St. Louis Rev. 153 (2021) [hereinafter Schär, *Decentralised Finance*].

³Douglas W. Arner, Janos Barberis & Ross P. Buckley, *FinTech, RegTech, and the Reconceptualization of Financial Regulation*, 37 Nw. J. Int'l L. & Bus. 1 (2016) [hereinafter Arner et al.].

⁴Katharina Pistor, *The Code of Capital: How the Law Creates Wealth and Inequality* (2019) [hereinafter Pistor]; Hyman P. Minsky, *Stabilizing an Unstable Economy* (1986) [hereinafter Minsky].

⁵Campbell R. Harvey, Ashwin Ramachandran & Joey Santoro, *DeFi and the Future of Finance* (2021) [hereinafter Harvey et al.].

⁶Matthias Lehmann, *Digital Assets in the Conflict of Laws: A Comparative Search for the Ideal Rule*, 2024 Sing. J. Legal Stud. 1, 2-5 [hereinafter Lehmann, *Digital Assets*]; Andrew Dickinson, *Cryptocurrencies and the Conflict of Laws*, in *Cryptocurrencies in Public and Private Law 1* (David Fox & Sarah Green eds., 2019).

structures outside its effective regulatory core.⁷ The result is an emerging asymmetry between increasingly sophisticated systems of automated financial governance and legal frameworks that remain territorially and institutionally grounded.

This article advances three interconnected claims. The first is descriptive and institutional: that Bitcoin-collateralised lending does not merely introduce new financial instruments, but constitutes a structural reorganisation of the functions traditionally performed by secured credit law—collateral creation, risk allocation, and enforcement—relocating them from identifiable legal institutions into protocol-based systems that operate without territorial anchoring, without identifiable intermediaries, and without the discretionary intervention that legal frameworks have historically presupposed.

The second claim is critical and doctrinal: that existing regulatory and conflict-of-laws frameworks—including MiCA Regulation and the principal European PIL instruments—were designed for a financial architecture that decentralised collateral systems have structurally displaced, and that the resulting legal gap is not a matter of incomplete implementation but of categorical inadequacy.

The third claim is constructive and original: that coherent legal governance of decentralised collateral requires the development of a transnational conflicts architecture organised around three principles—*technological control* as the primary connecting factor for proprietary relationships, *functional equivalence* as the organising principle for regulatory attribution, and *layered governance* as the architectural model for coordinating protocol norms, domestic property law, and international PIL instruments.

These three principles, the article argues, provide a framework capable of addressing the proprietary, jurisdictional, and insolvency dimensions of decentralised digital collateral without requiring either the wholesale redesign of existing PIL methodology or the abandonment of the territorial legal order that gives private international law its institutional coherence.⁸

This article uses Bitcoin-collateralised lending as a case study through which to examine the transformation of transnational financial governance. Drawing on digital asset regulation, financial governance, and PIL, it argues that DeFi lending protocols do not eliminate financial risk, but redesign its allocation and enforcement. The article further contends that this redesign exposes structural limitations within existing conflict-of-laws frameworks, particularly regarding the legal characterisation of digital collateral, the applicable law governing proprietary effects, and the enforceability of smart-contract-based liquidation mechanisms across jurisdictions.⁹

⁷Regulation (EU) 2023/1114 of the European Parliament and of the Council of 31 May 2023 on Markets in Crypto-Assets, 2023 O.J. (L 150) 1 [hereinafter MiCA]; Dirk A. Zetsche, Douglas W. Arner & Ross P. Buckley, Decentralized Finance, 6 J. Fin. Regulation 172 (2020) [hereinafter Zetsche et al., *Decentralized Finance*].

⁸Symeon C. Symeonides, *Choice of Law* 3-12 (2d ed. 2022); Lehmann, *supra* note 6; On the concept of categorical inadequacy in legal responses to technological disruption, see Rolf H. Weber, Regulatory Frameworks and Digital Financial Services: From FinTech to DeFi, 11 J. Fin. Regulation 1, 4-8 (2025) [hereinafter Weber, *Regulatory Frameworks*]; Rolf H. Weber, Governing DeFi: Regulatory Challenges and Policy Responses, 14 Eur. J. Risk Reg. 45, 46-52 (2023) [hereinafter Weber, *Governing DeFi*].

⁹UNIDROIT Principles on Digital Assets and Private Law (PRINS) (2023), *available at* <https://www.unidroit.org/instruments/digital-assets/>; Lehmann, *Digital Assets*, *supra* note 6; Christiane Wendehorst, Proprietary Rights in Digital Assets and the Conflict of Laws, in *Blockchain and Private International Law* 101 (Andrea Bonomi, Matthias Lehmann & Shaheez Lalani eds., 2023) [hereinafter Wendehorst, *Proprietary Rights*].

II. FROM INSTITUTIONAL TRUST TO VERIFIABLE COLLATERAL

Decentralised collateralised lending restructures the foundations upon which modern credit systems have historically operated. In conventional finance, credit relationships depend upon institutional trust, intermediary supervision, and territorially grounded systems of legal enforcement. Creditworthiness is therefore inseparable from institutional intermediation and from legal frameworks capable of attributing responsibility to identifiable actors operating within a determinate jurisdictional order.¹⁰

Bitcoin-collateralised lending alters this architecture fundamentally. In decentralised finance, risk allocation depends less on institutional judgment than on verifiable collateral, automated execution, and continuous protocol monitoring. The resulting shift reflects a deeper transformation in the governance of financial risk and in the mechanisms through which trust is produced within transnational markets.¹¹

In decentralised lending systems, collateral performs not merely a security function, but a governance function. Over-collateralisation, automated liquidation thresholds, and smart-contract enforcement collectively operate as mechanisms of endogenous discipline designed to preserve solvency without continuous human intervention. The protocol itself increasingly assumes functions traditionally exercised by banks, clearing institutions, custodians, and, to a certain extent, courts.¹²

Within this framework, Bitcoin occupies a distinctive position. Its liquidity, divisibility, transparency, and algorithmically constrained supply render it uniquely suited for collateralisation in decentralised environments. Unlike conventional collateral assets dependent upon institutional guarantees or territorial registration systems, Bitcoin collateral is verified cryptographically and enforced through protocol logic rather than discretionary institutional processes.¹³

Bitcoin-collateralised lending reveals a broader transition from institution-based credit toward collateral-based protocol governance. The central question is no longer whether borrowers are institutionally trustworthy, but whether collateral structures remain sufficiently resilient under conditions of volatility, automated enforcement, and cross-border legal uncertainty.¹⁴

Yet algorithmic discipline does not eliminate fragility. The stability of decentralised collateral systems ultimately depends upon the interaction between asset volatility, liquidity conditions, governance structures, and legal enforceability. The displacement of discretion by automation therefore reconfigures risk rather than removing it. Bitcoin-collateralised lending should accordingly be understood as part of a broader reconfiguration of transnational financial governance in which code, collateral, and distributed infrastructures increasingly perform functions historically exercised by legal institutions and regulated intermediaries.

¹⁰Pistor, *supra* note 4; Minsky, *supra* note 4; Arner et al., *supra* note 3.

¹¹ Primavera De Filippi & Aaron Wright, *Blockchain and the Law: The Rule of Code* (2018) [hereinafter De Filippi & Wright]; Kevin Werbach, *The Blockchain and the New Architecture of Trust* (2018) [hereinafter Werbach]; Zetzsche et al., *Decentralized Finance*, *supra* note 7.

¹² Satoshi Nakamoto, *Bitcoin: A Peer-to-Peer Electronic Cash System* (2008), <https://bitcoin.org/bitcoin.pdf>; Arvind Narayanan et al., *Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction* (2018); Andreas M. Antonopoulos, *Mastering Bitcoin: Programming the open blockchain* (2023).

¹³ Harvey et al., *supra* note 5.

¹⁴ Schär, *Decentralised Finance*, *supra* note 2.

III. COLLATERAL GOVERNANCE AND ALGORITHMIC ENFORCEMENT

Decentralised lending systems do not merely introduce new financial instruments; they reorganise the fundamental relationship between credit, collateral, and enforcement. In conventional finance, the stability of lending relationships depends upon institutional intermediation: banks, clearinghouses, custodians, and courts perform the functions of risk assessment, margin management, and enforcement through processes that are legally attributable, territorially situated, and subject to regulatory oversight. In decentralised finance, by contrast, these functions are increasingly performed through predefined operational rules embedded within protocol architecture. The legal question this displacement generates is not whether such systems are financially efficient, but whether the substitution of institutional agency by algorithmic execution is legally intelligible within existing frameworks of secured transactions law, regulatory oversight, and private international law.¹⁵

The central operational mechanism through which decentralised lending achieves stability is over-collateralisation. Borrowers are required to pledge digital assets whose market value substantially exceeds the amount borrowed—typically at maximum loan-to-value (LTV) ratios between 50% and 75%, depending on the protocol and the specific asset class. On Aave v3, for instance, wrapped Bitcoin operates at a maximum LTV of approximately 73%, with a liquidation threshold set at 78%; on MakerDAO (now Sky Protocol), DAI-generating vaults backed by Bitcoin have historically required collateralisation ratios of 150% or above. The LTV ratio consequently operates not merely as a financial indicator but as a mechanism of endogenous governance: it defines the conditions under which collateral is deemed sufficient, determines the trigger for automated enforcement, and delimits the protocol's exposure to solvency risk without requiring any discretionary human judgment at the moment of execution.¹⁶

This structure is stress-tested by automated liquidation mechanisms. Where the market value of pledged collateral falls below the liquidation threshold, the protocol initiates a liquidation process in which the collateral is offered at a discount to third-party liquidators, who restore the required collateral balance and receive a liquidation bonus in return. The significance of this mechanism lies not only in its technical efficiency but in the fundamental relocation of enforcement it produces: collateral realisation no longer depends upon judicial intervention, institutional discretion, or the exercise of contractual rights under domestic law, but upon the execution of protocol logic that operates continuously, globally, and without reference to the legal status of the borrower or the jurisdictional affiliation of any party to the transaction.

The operational consequences of this design are not merely theoretical. On 12-13 March 2020—a period known in DeFi markets as 'Black Thursday'—a severe and rapid decline in the price of Ether generated cascading liquidations across the MakerDAO system, producing a collateral deficit of approximately \$5.4 million in undercollateralised vaults and, critically, a series of zero-bid auction outcomes in which liquidators acquired collateral for near-zero consideration. The

¹⁵ Schär, *Decentralised Finance*, supra note 2, at 155-60; Pistor, supra note 4, at 1-18; Harvey et al., supra note 5, at 15-28; Robert C. Merton, *A Functional Perspective of Financial Intermediation*, 24 *Fin. Mgmt.* 23, 25-30 (1995) [hereinafter Merton].

¹⁶ Aave Protocol v3, *Risk Parameters Documentation* (2024), <https://docs.aave.com/risk/asset-risk/risk-parameters>, last visited 25 May 2026; MakerDAO Collateral Risk Framework (2023), <https://makerburn.com>, last visited 25 May 2026; Sam Werner et al., *SoK: Decentralized Finance*, in *Proceedings of the 4th ACM Conference on Advances in Financial Technologies* 30, 38-45 (2022) [hereinafter Werner et al., *SoK: DeFi*].

event exposed a structural tension between the protocol's automated enforcement logic and the reasonable expectations of borrowers—a tension that existing contract law and secured transactions frameworks were entirely unprepared to address.¹⁷

Smart contracts play a constitutive role in this architecture. They do not merely execute agreements; in a meaningful functional sense they *are* the agreement—the terms of the lending relationship, including collateral requirements, interest rates, and liquidation parameters, are encoded in contract bytecode and enforced autonomously upon the occurrence of specified on-chain conditions. This conflation of legal obligation and technical execution creates a fundamental challenge for legal characterisation. The question of whether a smart-contract-based lending arrangement constitutes a secured loan, a title transfer with a repurchase right, a conditional sale, or a *sui generis* digital financial instrument cannot be resolved by reference to the parties' subjective intentions, because the contract leaves no room for interpretive discretion: execution is automatic, deterministic, and irreversible. This is precisely the quality that renders smart contracts operationally attractive and, simultaneously, legally intractable.¹⁸

Custody acquires particular analytical significance in this context, and its legal treatment is considerably more complex than in conventional secured transactions. Three distinct custody arrangements are operative in decentralised lending systems. In *non-custodial* arrangements, the borrower retains control of their private key throughout the lending relationship, and the smart contract functions as a protocol-level lien without effecting a transfer of possession in any conventional legal sense. In *protocol-custodial* arrangements, the borrower transfers the asset to a smart-contract address controlled by the protocol's governance structure, effecting a transfer of technical control without a transfer to any identifiable legal person or institution. In *hybrid* arrangements—increasingly common in institutional DeFi—multisignature mechanisms distribute control across a combination of protocol logic, designated key-holders, and governance token votes. The distinction between these three arrangements has direct and unresolved implications for insolvency treatment, the characterisation of proprietary rights under domestic law, and the determination of the applicable law under PIL frameworks.¹⁹

The legal characterisation of the automated liquidation event itself deserves sustained attention, because it sits at the intersection of contract law, property law, and enforcement procedure in a way that existing frameworks cannot accommodate. When a liquidation is triggered, the following sequence occurs automatically and irreversibly: the protocol identifies that the collateral value has fallen below the liquidation threshold; it accepts a bid from a third-party liquidator; it transfers the collateral to the liquidator's wallet address; and it settles the outstanding loan balance, retaining a protocol fee. No court order is obtained. No notice is served upon the borrower. No domestic enforcement procedure is followed.

¹⁷ Cyrus Younessi, *The Market Collapse of March 12, 2020: How It Impacted MakerDAO*, MakerDAO Blog (Mar. 19, 2020), <https://blog.makerdao.com/the-market-collapse-of-march-12-2020-how-it-impacted-makerdao/>; Werner et al., SoK: DeFi, *supra* note 16, at 55-58; Harvey et al., *supra* note 15, at 95-102.

¹⁸ De Filippi & Wright, *supra* note 11, at 72-85; Max Raskin, *The Law and Legality of Smart Contracts*, 1 *Geo. L. Tech. Rev.* 305, 315-22 (2017) [hereinafter Raskin]; Jason G. Allen et al., *Legal and Regulatory Considerations for Digital Assets*, CCAF Publications 1, 18-26 (2020) [hereinafter Allen et al.].

¹⁹ Allen et al., *supra* note 18, at 30-38; Tsz Yin Chan, *The Nature of Property in Cryptoassets*, 43 *Legal Stud.* 480, 492-500 (2023) [hereinafter Chan]; Wendehorst, *Proprietary Rights*, *supra* note 9; UNIDROIT Principles on Digital Assets and Private Law (PRINS) art. 5, Explanatory Notes paras. 5.3-5.7 (2023).

The entire sequence may execute within a single blockchain block—in seconds—entirely on-chain and without any possibility of intermediate legal intervention. Whether this sequence constitutes extrajudicial enforcement, self-help repossession, an automatic condition subsequent, or a proprietary reallocation that produces legally cognisable title in the liquidator is unresolved across all major legal systems. It is nonetheless a question with direct consequences for the rights of borrowers in distress, the obligations of liquidators, the treatment of residual collateral value in insolvency, and the cross-border enforceability of the resulting on-chain title transfer.²⁰

What decentralised collateral governance achieves, in summary, is not the elimination of financial risk but its systematic redistribution and technical recharacterisation. Risk is transferred from the lending institution—which, in conventional finance, bears credit risk, operational risk, and systemic risk in exchange for regulatory oversight and accountability—to the borrower, who bears liquidation risk and collateral volatility risk; to the liquidator, who bears execution and slippage risk; and, residually, to the protocol's governance token holders, who bear tail risk in the event of undercollateralisation that exceeds the protocol's reserve capacity.

Under normal market conditions, this redistribution produces a financially resilient and operationally efficient system. Under conditions of acute market stress—as demonstrated by the March 2020 MakerDAO event, the May 2022 Terra/Luna algorithmic stablecoin collapse, and the subsequent cascading liquidations across interconnected DeFi protocols—the same mechanisms that ensure stability amplify price declines, generate acute liquidity crises, and produce legal consequences for which no adequate framework of accountability currently exists. In this sense, DeFi collateral governance exhibits the structural characteristic that MINSKY identified as endemic to leverage-based systems: stability is endogenously destabilising, and the very mechanisms designed to contain risk become, under sufficient stress, the vectors through which systemic fragility is transmitted.²¹

The governance architecture of decentralised collateral systems thus presents legal scholarship with a challenge that is simultaneously technical, doctrinal, and systemic. It is technical because adequate legal analysis requires an understanding of how LTV ratios, liquidation thresholds, smart-contract bytecode, and on-chain settlement actually function. It is doctrinal because the absence of legal characterisation for automated liquidation, protocol custody, and smart-contract enforcement creates genuine uncertainty across contract law, property law, secured transactions law, and insolvency law simultaneously. It is systemic because the cross-border operation of decentralised protocols means that these doctrinal uncertainties arise concurrently across multiple jurisdictions, generating conflicts of laws that PIL frameworks were not designed and are not

²⁰ Schär, *Decentralised Finance*, supra note 15, at 163-66; Werner et al., *SoK: DeFi*, supra note 16, at 45-52; Philipp Paech, *The Governance of Blockchain Financial Networks*, 80 *Mod. L. Rev.* 1073, 1083-90 (2017) [hereinafter Paech, *Governance*]; De Filippi & Wright, supra note 11, at 115-25. On the absence of procedural protections in on-chain enforcement and the implications for domestic law: United Kingdom Law Commission, *Digital Assets: Final Report*, Law Com. No. 412, at 45-52 (2023).

²¹ On the Terra/Luna collapse and DeFi contagion: Financial Stability Board, *Global Regulatory Framework for Crypto-Asset Activities: Final Report* 15-22 (July 2023) [hereinafter FSB, *Global Regulatory Framework*]; Dirk A. Zetzsche, Ross P. Buckley, Douglas W. Arner & Maurits van Ek, *Remaining Regulatory Challenges in Digital Finance and Crypto-Assets after MiCA 1*, 5-10 (Eur. Parliament, ECON Comm., May 2023), <https://ssrn.com/abstract=4487516> [hereinafter Zetzsche et al., *Remaining Challenges*]. On the Minsky dynamic applied to DeFi: Hyman P. Minsky, *Stabilizing an Unstable Economy* 195-210 (1986); Werner et al., *SoK: DeFi*, supra note 16, at 58-64.

currently equipped to resolve. It is to these regulatory and private international law dimensions that the analysis now turns.²²

IV. MICA AND THE REGULATORY PERIMETER OF DEFI

MiCA represents the most comprehensive regulatory framework yet adopted for crypto-asset markets within the European Union. Its significance lies not merely in the harmonisation of market access rules for crypto-assets and service providers, but in the attempt to integrate parts of the digital asset economy into the institutional logic of financial regulation.²³

This tension is particularly visible in the context of decentralised collateralised lending. MiCA regulates crypto-asset service providers (CASPs), imposing requirements concerning authorisation, governance, transparency, custody, and consumer protection. However, where lending relationships are organised through smart contracts operating without centralised control, the connection between regulatory obligation and legal responsibility becomes substantially more difficult to establish.²⁴

The resulting difficulty is not merely technical, but conceptual. Traditional financial regulation presupposes the existence of legal subjects capable of bearing responsibility for the management of risk, compliance, and enforcement. Decentralised finance destabilises these assumptions by distributing operational functions across automated systems that do not correspond easily to conventional institutional categories.²⁵

MiCA partially recognises this structural problem. Recital 22 excludes fully decentralised crypto-asset services from the regulation's scope where no intermediary can be identified. Yet this exclusion simultaneously reveals the limits of the current regulatory model. The more decentralised the system becomes, the more difficult it is to integrate within regulatory frameworks designed around institution-based supervision and territorially attributable conduct.²⁶

The emergence of decentralised collateral systems consequently exposes a growing asymmetry between automated forms of transnational financial coordination and legal frameworks that remain structurally tied to territorial authority and intermediary-based supervision. MiCA undoubtedly represents an important step toward regulatory coherence in crypto asset markets, yet its intermediary-based structure remains conceptually insufficient for fully decentralised environments: whether legal systems historically organised around identifiable institutions and territorially attributable conduct remain capable of governing increasingly automated and geographically diffuse forms of economic ordering.²⁷

V. DIGITAL COLLATERAL AND THE COLLAPSE OF TERRITORIAL CONNECTING FACTORS

²² Zetzsche et al., *Decentralized Finance*, supra note 7, at 195-202; De Filippi & Wright, supra note 11, at 195-205; Weber, *Governing DeFi*, supra note 8, at 48-55.

²³ MiCA, supra note 7; Zetzsche et al., *Decentralized Finance*, supra note 7.

²⁴ De Filippi & Wright, supra note 11; Werbach, supra note 11.

²⁵ Werbach, supra note 11; Zetzsche et al., *Decentralized Finance*, supra note 7.

²⁶ MiCA, supra note 7, recital 22.

²⁷ Zetzsche et al., *Remaining Challenges*, supra note 21.

The emergence of decentralised collateral systems raises one of the most fundamental challenges currently confronting Private International law. Modern conflict-of-laws frameworks were constructed upon the assumption that legally significant relationships could ultimately be anchored to territorially identifiable connecting factors: the domicile of the parties, the place of performance, the *situs* of assets, or the location of an intermediary capable of exercising legal control over the subject matter of the transaction. Digital collateral in decentralised finance destabilises each of these assumptions simultaneously, exposing a structural tension between the territorial architecture of classical PIL methodology and the operational logic of distributed financial systems.²⁸

In conventional secured transactions, the proprietary effects of collateral arrangements are governed by the law of the jurisdiction in which the asset is situated at the time of the relevant transaction—the *lex situs* rule. This connecting factor performs a dual function: it provides a determinate and predictable choice-of-law rule for third parties and insolvency administrators, and it ensures that proprietary rights are enforceable within a legally organised territory. Decentralised digital assets do not fit within this structure. Bitcoin collateral may be pledged through distributed infrastructures operating simultaneously across multiple jurisdictions without any stable geographical anchor, identifiable intermediary, or territorially situated registry capable of performing the coordinating function that the *lex situs* rule presupposes.²⁹

The inadequacy of the *lex situs* rule in digital asset contexts has been extensively documented in recent comparative scholarship. The classical rule, as applied through the general principles of private international law in most European jurisdictions and as reflected in the PRIMA rule (Place of the Relevant Intermediary Approach) for intermediated securities, locates proprietary rights by reference to the physical or notional location of the asset, or by reference to the account-holding institution. Neither approach translates adequately to decentralised environments in which assets exist as entries across distributed ledger infrastructure, validators operate across multiple jurisdictions, and no central intermediary holds a legally determinative position.³⁰

The conflict-of-laws implications become particularly acute in cross-border lending relationships. A borrower may be domiciled in one jurisdiction, access liquidity through a protocol governed by token-based voting mechanisms dispersed globally, pledge collateral verified across distributed ledger infrastructure operating in multiple countries, and interact through a front-end interface operated from yet another jurisdiction. Under such conditions, the identification of the applicable law governing proprietary effects, enforcement rights, or insolvency treatment becomes analytically indeterminate under classical connecting factor methodology. Recent national

²⁸ Lehmann, *supra* note 6; Andrew Dickinson, Cryptocurrencies and the Conflict of Laws, in *Cryptocurrencies in Public and Private Law* 1, 3–5 (David Fox & Sarah Green eds., 2019).

²⁹ UNIDROIT Principles on Digital Assets and Private Law (PRINS) arts. 2, 5, Explanatory Notes paras. 2.1-2.4 (2023), available at <https://www.unidroit.org/instruments/digital-assets/> [hereinafter UNIDROIT PRINS]; Wendehorst, Proprietary Rights, *supra* note 9, at 103-10.

³⁰ Burcu Yüksel Ripley & Florian Heindler, The Law Applicable to Crypto Assets: What Policy Choices Are Ahead of Us?, in *Blockchain and Private International Law*, at 261-65; Amy Held, Crypto Assets and Decentralised Ledgers: Does Situs Actually Matter?, in *Blockchain and Private International Law*, at 257, 258-60; Koji Takahashi, Law Applicable to Proprietary Issues of Crypto-Assets, 18 J. Priv. Int'l L. 339, 341-50 (2022) [hereinafter Takahashi].

legislative initiatives illustrate the divergence of responses in the absence of international coordination.³¹

Comparative legal systems have produced divergent and, as yet, only partially satisfactory responses to this challenge. The United Kingdom Law Commission's *Digital Assets: Final Report* (2023) recommends recognition of a third category of personal property for digital assets—distinct from things in possession and things in action—acknowledging that existing property law categories are structurally inadequate to accommodate cryptographic assets. Liechtenstein's Token and TT Service Provider Act (*TVTG*, 2019) addresses the problem through the Token Container Model, treating the token as the legally operative carrier of rights. Germany's Electronic Securities Act (*eWpG*, 2021) extends the concept of legal possession to entries in electronic securities registers. Singapore has aligned its framework with the UNIDROIT PRINS, anchoring proprietary analysis to 'control' as a technologically neutral legal concept. These divergent national approaches, while innovative within their respective domestic orders, collectively illustrate the fragmentation that arises in the absence of transnational PIL coordination.³²

The practical consequences of this fragmentation are not merely theoretical. The Chapter 11 proceedings of Celsius Network LLC before the United States Bankruptcy Court for the Southern District of New York provide a concrete illustration of the proprietary uncertainties generated by decentralised digital asset structures. In those proceedings, the central question—whether digital assets deposited in yield-bearing accounts constituted property of the bankruptcy estate or remained the property of depositors—generated novel and unresolved questions precisely at the intersection of property law characterisation and cross-border enforcement. The Celsius case demonstrates that the absence of coherent PIL rules for digital collateral creates not only doctrinal uncertainty, but direct and irreversible harm for creditors in multi-jurisdictional insolvency proceedings.³³

The most analytically promising response to these challenges lies in the progressive displacement of territorial location by *control* as the operative legal concept for digital asset transactions. The UNIDROIT Principles on Digital Assets and Private Law (PRINS, 2023) provide the most systematic articulation of this approach, defining control as the practical ability to derive the benefit of a digital asset and to prevent others from doing so. Under this framework, the legally relevant question is not where the asset is geographically located, but who exercises effective control over the cryptographic key or mechanism through which the asset may be transferred or encumbered. This shift reflects a fundamental reconceptualisation of the relationship between legal

³¹ Lehmann, *supra* note 6, at 8–14; Symeon C. Symeonides, *Choice of Law* 427–32 (2d ed. 2022); Louise Gullifer & Orkun Akseli, Introduction, in *Secured Transactions Law Reform: Principles, Policies and Practice* 1, 12–18 (Louise Gullifer & Orkun Akseli eds., 2016).

³² United Kingdom Law Commission, *Digital Assets: Final Report*, Law Com. No. 412, at 12–18, 45–52 (2023) [hereinafter UK Law Commission, *Digital Assets Final Report*]; Token- und VT-Dienstleister-Gesetz [TVTG] [Token and TT Service Provider Act] of Oct. 3, 2019, LGBL 2019 Nr. 301 (Liech.); Gesetz zur Einführung von elektronischen Wertpapieren [eWpG] [Electronic Securities Act] of June 3, 2021, BGBl. I at 1423 (Ger.); Monetary Authority of Singapore Act (Cap. 186A) (amended 2021) (Sing.); Law Commission, *Digital Assets and Electronic Trade Documents in Private International Law: Consultation Paper* (2025), <https://lawcom.gov.uk/project/digital-assets-and-electronic-trade-documents-in-private-international-law/>.

³³ In re Celsius Network LLC, No. 22-10964 (MG) (Bankr. S.D.N.Y. July 13, 2022) (Chapter 11 petition); In re Celsius Network LLC, No. 22-10964 (MG), 2023 WL 3671029 (Bankr. S.D.N.Y. May 25, 2023).

authority and technical infrastructure that existing PIL methodologies have not yet fully incorporated.³⁴

This reconceptualisation creates further complications at the level of protocol governance. In decentralised lending systems, governance is exercised through token-based voting mechanisms in which authority is diffused across a geographically dispersed community of stakeholders. The rules governing collateral thresholds, liquidation parameters, and protocol modifications are determined by decentralised governance proposals rather than by identifiable legal subjects acting within a determinate jurisdictional order. The resulting system produces what has been aptly described as a form of *lex cryptographia*—a system of norms embedded in code and enforced algorithmically, which operates in parallel with, and frequently in tension with, territorially organised legal systems. PIL frameworks built around identifiable parties, attributable acts, and judicially enforceable obligations are structurally ill-suited to govern disputes arising within such systems.³⁵

The limitations of existing European PIL instruments are correspondingly evident. The Rome I Regulation provides choice-of-law rules for contractual obligations, operating on the assumption that a concluded contract can be identified and attributed to parties acting within a recognisable legal framework. The Rome II Regulation governs non-contractual obligations arising from legally attributable acts or events. The Brussels I *bis* Regulation allocates jurisdiction by reference to the defendant's domicile or the place of the harmful event. None of these instruments provides adequate guidance for disputes arising from the automated liquidation of digital collateral in a protocol that lacks a legal seat, operates across distributed infrastructure, and executes obligations through algorithmic rather than volitional performance.³⁶

These structural limitations carry systemic implications beyond the doctrinal. The Financial Stability Board's 2023 Global Regulatory Framework for Crypto-Asset Activities identifies regulatory fragmentation as one of the principal risks associated with decentralised finance, noting that divergent national approaches create opportunities for regulatory arbitrage, increase the risk of asset stranding in insolvency, and undermine the coherence of consumer protection measures across jurisdictions. The cross-border dimensions of DeFi collateral systems therefore present not only doctrinal challenges for PIL scholarship, but governance challenges for international

³⁴ UNIDROIT PRINS, *supra* note 29, arts. 5-7, Explanatory Notes paras. 5.1-5.12 (introducing 'control' as the operative legal concept for digital assets, displacing possession-based frameworks); Lehmann, Digital Assets, *supra* note 6, at 15-22 (advocating for a 'control-based connecting factor' as the primary rule for determining the applicable law governing proprietary issues in digital asset transactions); Wendehorst, Proprietary Rights, *supra* note 9, at 115-22.

³⁵ Paech, Governance, *supra* note 20, at 1078-90; De Filippi & Wright, *supra* note 11, at 78-85 (coining the term *lex cryptographia* to describe algorithmically enforced normative systems operating outside territorial legal orders).

³⁶ Council Regulation (EC) No 593/2008 of 17 June 2008 on the Law Applicable to Contractual Obligations (Rome I), arts. 3-4, 2008 O.J. (L 177) 6; Regulation (EC) No 864/2007 of the European Parliament and of the Council of 11 July 2007 on the Law Applicable to Non-Contractual Obligations (Rome II), art. 4, 2007 O.J. (L 199) 40; Regulation (EU) No 1215/2012 of 12 December 2012 on Jurisdiction and the Recognition and Enforcement of Judgments in Civil and Commercial Matters (Brussels I *bis*), arts. 4, 7, 2012 O.J. (L 351) 1; Dickinson, *supra* note 6, at 22-30.

regulatory coordination that neither MiCA nor any existing PIL instrument is currently equipped to resolve.³⁷

Digital collateral in decentralised finance thus exposes the limits of a conflict-of-laws methodology historically organised around three interlocking assumptions: that legal relationships involve identifiable parties; that legally significant acts occur at a determinate place; and that proprietary rights attach to assets with a fixed and ascertainable location. Decentralised collateral systems displace all three. The lender may be an anonymous wallet address, the borrower a pseudonymous actor, the enforcing mechanism an algorithm with no discernible jurisdictional seat, and the collateral a cryptographic token distributed across thousands of nodes worldwide. The collapse of stable territorial connecting factors does not imply the disappearance of legal order, but rather the urgent necessity of constructing forms of transnational legal coordination capable of addressing decentralised systems whose operational logic no longer corresponds to the territorial assumptions upon which classical PIL was built.³⁸

The challenge is not to abandon the PIL framework but to reconstruct its methodology around the functional realities of decentralised financial governance—moving from territorial localisation toward technologically mediated control as the principal organising concept for proprietary conflicts involving digital collateral.³⁹

VI. TOWARD A TRANSNATIONAL CONFLICTS ARCHITECTURE FOR DIGITAL COLLATERAL

The preceding analysis has established that decentralised digital collateral cannot be governed adequately by any existing PIL instrument — not because those instruments are poorly drafted, but because they were designed for a world in which legally significant relationships could be anchored to territory, attributed to identifiable persons, and enforced through institutional mechanisms capable of exercising coercive authority. That world has not disappeared; but it now coexists with a parallel financial infrastructure whose operational logic is fundamentally incompatible with territorial localisation as an organising principle. The question is no longer whether existing frameworks are sufficient. They are not. The question is what must be built in their place — or, more precisely, alongside them.⁴⁰

Decentralised finance does not merely introduce new financial instruments into existing legal frameworks; it constitutes what DE FILIPPI and WRIGHT have described as a system of *lex cryptographia*—a normative order embedded in code, enforced algorithmically, and operating across distributed infrastructures without reference to any territorially bounded legal authority. The implications of this observation for PIL are profound. A legal system organised around the attribution of rights and obligations to identifiable subjects, located within determinable

³⁷ FSB, Global Regulatory Framework, *supra* note 21, at 18-26; Bank for International Settlements, Annual Economic Report 2022, Ch. II: The Future Monetary System 72-80 (2022); Zetzsche et al., Remaining Challenges, *supra* note 21, at 8-15.

³⁸ Takahashi, *supra* note 30, at 352-62; Wendehorst, Proprietary Rights, *supra* note 9, at 118-25; Lehmann, Digital Assets, *supra* note 6, at 24-30.

³⁹ UNIDROIT PRINS, *supra* note 29, Explanatory Notes to art. 5, paras. 5.9-5.14; Lehmann, Digital Assets, *supra* note 6, at 28-35.

⁴⁰ Lehmann, *supra* note 6, at 28-35; Rolf H. Weber, Digital Finance: Social Networks, Mobile Payments and Electronic Commerce 112-18 (2014).

jurisdictions, and enforced through institutional mechanisms capable of exercising coercive authority, encounters a systematic governance challenge when the subject of the relationship is a borderless cryptographic token, the governing mechanism is an autonomous protocol, and the enforcing agent is an algorithm.⁴¹

The first and most fundamental pillar of a coherent transnational conflicts architecture for digital collateral is the adoption of *technological control* as the primary connecting factor for proprietary relationships involving digital assets. The UNIDROIT PRINS provide the most developed articulation of this concept, defining control as the practical ability to derive the benefit of a digital asset and to prevent others from doing so. This definition is deliberately technology-neutral: it does not presuppose any particular architecture, protocol, or implementation standard. Its legal significance lies in the displacement of territorial possession and institutional intermediation as the operative criteria for determining the allocation of proprietary rights.⁴²

Applied to decentralised collateral structures, the control-based framework produces a coherent analytical sequence. At the moment of collateral creation, the relevant legal question is not where the asset is geographically situated, but who exercises control over the cryptographic mechanism through which the asset can be transferred or encumbered. Where collateral is locked in a smart contract, control passes from the borrower to the protocol—or, more precisely, to the set of governance rules embedded in the contract—at the moment of deposit. The applicable law governing the proprietary effects of that transfer should therefore be determined by reference to the law most closely connected to the control mechanism: in practice, the jurisdiction under whose law the smart contract was deployed, the validators who execute it principally operate, or—as a default of last resort—the law designated by the protocol's governance documentation.⁴³

The second pillar of the proposed architecture concerns the principle of *functional equivalence* in transnational regulatory coordination. Drawing on WEBER's analysis of governance design for digital financial systems, this principle holds that the legal treatment of a given function should be determined by the function itself rather than by the institutional form through which it is exercised. Where a smart contract performs functions equivalent to those of a secured creditor, a custodian, or a financial intermediary, the legal regime applicable to that function should be determined by reference to its functional characteristics rather than its technical architecture. This approach avoids the twin risks of over-regulation—which would stifle legitimate innovation by forcing decentralised systems into frameworks designed for institutional actors—and under-regulation—which would permit decentralised systems to evade legal accountability by exploiting the absence of identifiable intermediaries.⁴⁴

⁴¹ De Filippi & Wright, *supra* note 11, at 195-203; Primavera De Filippi & Sinclair Davidson, *Blockchain Abstraction: The Political and Economic Consequences of Coding Trust*, 35 Computer L. & Security Rev. 1, 4-8 (2019).

⁴² UNIDROIT PRINS, *supra* note 29, arts. 5-7, Explanatory Notes paras. 5.9-5.14; Lehmann, *Digital Assets*, *supra* note 6, at 15-25; Wendehorst, *Proprietary Rights*, *supra* note 9, at 118-25.

⁴³ UNIDROIT PRINS, *supra* note 29, art. 5(1) ('A person has control of a digital asset if that person: (a) has the ability to prevent others from obtaining control of the digital asset; and (b) has the exclusive ability to: (i) transfer or encumber the digital asset; or (ii) authorise another person to do so'). This definition is deliberately technology-neutral: it does not presuppose blockchain, private keys, or any specific technical implementation, thereby preserving the rule's durability across successive generations of digital asset infrastructure.

⁴⁴ Symeon C. Symeonides, *Choice of Law* 432-40 (2d ed. 2022); Weber, *Regulatory Frameworks*, *supra* note 8, at 12-18; Lehmann, *supra* note 6, at 30-38.

The functional equivalence principle has immediate consequences for the relationship between MiCA and decentralised collateral systems. MiCA establishes a regulatory perimeter around identifiable crypto-asset service providers but excludes fully decentralised services from its scope where no intermediary can be identified. This exclusion is operationally coherent given MiCA's institutional logic, but it creates a regulatory gap that the functional equivalence principle can help to address. Where a decentralised protocol performs functions that, if performed by a natural or legal person, would fall within the scope of MiCA's CASP regime—including custody, collateralisation, and automated enforcement—the functional equivalence principle supports the application of analogous regulatory standards to the extent technically feasible. This does not require the identification of a responsible legal person; it requires, rather, the development of protocol-level compliance standards capable of being embedded in the governance architecture of decentralised systems.⁴⁵

The third pillar concerns the legal recognition of protocol governance as a form of normative ordering capable of producing legally relevant effects. Current PIL methodology treats governance decisions within decentralised protocols—parameter changes, liquidation threshold modifications, emergency halts—either as irrelevant technical events or, at best, as contractual modifications subject to the law of the underlying transaction. Neither characterisation is adequate. As PAECH has argued, the governance of blockchain financial networks constitutes a quasi-constitutional order with its own amendment procedures, separation of functions, and enforcement mechanisms. Recognising this structural reality requires PIL to develop new categories capable of characterising governance decisions not merely as contractual acts, but as rule-making events with proprietary and regulatory implications capable of affecting third parties across jurisdictions.⁴⁶

A fourth dimension of the proposed architecture concerns the cross-border enforceability of automated liquidation events. In traditional secured transactions, the enforcement of security interests across jurisdictions is governed by principles of comity, reciprocal recognition, and the application of the *lex situs* at the time of enforcement. Automated liquidations in DeFi systems do not generate judicial or quasi-judicial acts that can be recognised or enforced in the conventional sense. They produce, rather, irreversible on-chain state changes—the transfer of collateral to a liquidator—that are legally ambiguous from the perspective of most domestic legal systems. The critical question is whether such on-chain state changes should be treated as self-executing transfers of title, requiring no further domestic recognition; as events generating claims in restitution or unjust enrichment; or as extrajudicial enforcement acts requiring post-hoc judicial scrutiny. The answer will differ across jurisdictions and will depend critically on how the underlying collateral relationship is characterised.⁴⁷

Drawing together the foregoing analysis, this article proposes a three-tier conflicts rule for digital collateral in decentralised lending systems. At the first tier, the proprietary effects of a digital collateral arrangement should be governed by the law expressly designated by the parties, or by the law chosen by protocol governance, provided that such designation is made in good faith and

⁴⁵ MiCA, *supra* note 7, recital 22, art. 2(4) (excluding fully decentralised crypto-asset services from scope); Zetzsche et al., *Remaining Challenges*, *supra* note 27, at 8–15.

⁴⁶ De Filippi & Wright, *supra* note 11, at 78–85 (*lex cryptographia*); Paech, *Governance*, *supra* note 20, at 1090–98; Weber, *Regulatory Frameworks*, *supra* note 8, at 18–24.

⁴⁷ Weber, *Governing DeFi*, *supra* note 8, at 52–60; FSB, *Global Regulatory Framework*, *supra* note 21, at 28–34; Zetzsche et al., *Decentralized Finance*, *supra* note 7, at 195–202.

does not operate to deprive a third party or insolvency administrator of rights that would otherwise be mandatory under the applicable law of the third party's jurisdiction. At the second tier, in the absence of an effective choice, the applicable law should be the law of the jurisdiction with which the control mechanism—defined by reference to the UNIDROIT PRINS concept of control—is most closely connected; in most cases, this will be the law of the jurisdiction where the principal validation infrastructure operates or where the protocol's governance is predominantly exercised. At the third tier, as a default of last resort applicable where neither tier one nor tier two yields a determinable result, the applicable law should be the law of the jurisdiction in which the collateral-taker (or its functional equivalent) has its principal place of business or habitual residence.⁴⁸

The implementation of such a framework cannot be achieved through unilateral domestic legislation alone. It requires, at minimum, a coordinated international response involving the UNIDROIT Secretariat, the European Law Institute, the Hague Conference on Private International Law, and the Financial Stability Board. The UNIDROIT PRINS provide a foundation, but they do not, as yet, address the specific conflicts implications of collateralisation and automated enforcement in decentralised systems with the granularity required for practical application. The ELI's ongoing work on digital assets, including the Digital Assets Working Group established in the context of the broader law reform agenda for digital markets, constitutes a critical institutional locus for the development of the proposed architecture within the European context.⁴⁹

The insolvency dimension of the proposed architecture deserves particular attention. Cross-border insolvency proceedings involving DeFi protocols—of which the Celsius Network proceedings are the most instructive example to date—have demonstrated that the absence of coherent PIL rules for digital collateral creates direct and irreversible harm. The question of whether digital assets pledged as collateral under a DeFi lending arrangement are property of the insolvency estate, property of the collateral-taker, or subject to automatic enforcement through the protocol mechanism cannot be answered adequately by reference to existing insolvency conflict-of-laws frameworks, which presuppose identifiable debtors, ascertainable estate assets, and territorial insolvency proceedings capable of producing universal or modified universal effects. The three-tier rule proposed above, if adopted into the applicable insolvency regulation—including through a targeted amendment to the EU Insolvency Regulation—would provide a workable framework for the determination of the applicable law governing the proprietary status of digital collateral in insolvency contexts.⁵⁰

The architecture proposed here is best understood not as a definitive codification, but as a framework of principles capable of guiding legal development over time. WEBER's concept of layered governance for transnational digital systems provides a useful analytical model: just as internet governance operates through a layered architecture in which technical standards,

⁴⁸ UNIDROIT PRINS, *supra* note 29, arts. 8-10 (addressing conflict-of-laws rules for digital asset transactions); Wendehorst, *Proprietary Rights*, *supra* note 9, at 122-30; Koji Takahashi, *supra* note 30. The proposal advanced here builds upon and extends these analyses by applying the control-based framework specifically to the collateralisation relationship in DeFi lending, a context that has not previously received systematic PIL treatment.

⁴⁹ Lehmann, *supra* note 6, at 33-40; Weber, *Governing DeFi*, *supra* note 8, at 55-62; European Law Institute, *Statement on the EU Commission's Proposal for a Regulation on Markets in Crypto-Assets 12-18 (2021)* [hereinafter *ELI Statement*]; Paech, *Governance*, *supra* note 20, at 1092-1100.

⁵⁰ UK Law Commission, *Digital Assets Final Report*, *supra* note 32, at 45-56; *In re Celsius Network LLC*, No. 22-10964 (MG), 2023 WL 3671029, at 8-12 (Bankr. S.D.N.Y. May 25, 2023); FSB, *Global Regulatory Framework*, *supra* note 21, at 30-38.

contractual relationships, and regulatory norms interact without any single layer having comprehensive authority, PIL for digital collateral should operate through a hierarchy of coordinated rules in which protocol-level governance, domestic property law, and international PIL instruments each play a defined and mutually reinforcing role. The key insight is that no single legal instrument can provide a comprehensive solution to the governance challenges posed by decentralised finance; the task is rather to construct a coherent architecture in which multiple sources of normativity interact without producing irresolvable conflicts.⁵¹

The development of a transnational conflicts architecture for digital collateral is not a marginal concern of technical legal scholarship. It is a condition of the legal intelligibility of an increasingly significant segment of global financial activity. Decentralised lending protocols already manage billions of dollars of digital asset collateral across jurisdictions, and their systemic importance is growing. The absence of coherent PIL rules for the proprietary, jurisdictional, and insolvency dimensions of decentralised collateral does not make those dimensions disappear; it makes them legally unpredictable, practically unenforceable, and systemically risky. The proposal advanced in this article—grounded in technological control as a connecting factor, functional equivalence as a regulatory principle, and layered governance as an architectural model—offers a framework through which legal scholarship can begin to close the gap between the operational reality of decentralised finance and the normative frameworks required to govern it.⁵²

VII. CONCLUSION

Decentralised collateral systems represent more than a technological evolution in financial markets. They constitute a structural reorganisation of the functions that legal systems have historically reserved for identifiable institutions: the assessment of credit risk, the custody of pledged assets, the management of margin, and the enforcement of security interests upon default. Bitcoin-collateralised lending protocols such as Aave and MakerDAO do not merely replicate these functions in digital form; they redistribute them across automated infrastructures that operate without territorial anchoring, without identifiable intermediaries, and without the procedural protections that domestic legal systems have developed over centuries of secured transactions law. The Black Thursday event of March 2020—in which MakerDAO's automated liquidation mechanisms produced zero-bid auction outcomes affecting borrowers across multiple jurisdictions—illustrates with precision what is at stake when legal frameworks fail to keep pace with the governance logic of decentralised finance.

This article has advanced three interconnected claims and sought to demonstrate each of them through doctrinal analysis. The first, descriptive claim—that DeFi lending constitutes a categorical displacement of institutional agency rather than a marginal extension of existing financial practice—is supported by the architecture of over-collateralisation, algorithmic enforcement, and protocol-based governance. The second, critical claim—that existing regulatory and PIL frameworks are categorically inadequate rather than merely incomplete—is borne out by the analysis of MiCA's structural exclusion of fully decentralised services and by the collapse of the

⁵¹ Weber, *Governing DeFi*, supra note 8, at 60–68; Rolf H. Weber, *Internet of Things: Legal Perspectives* 145–52 (2d ed. 2016) (developing the concept of ‘layered governance’ for transnational digital systems); De Filippi & Wright, supra note 11, at 200–08.

⁵² UNIDROIT PRINS, supra note 29, Preamble P.6–P.10; Lehmann, supra note 6, at 38–45; Weber, *Regulatory Frameworks*, supra note 8, at 22–28; Paech, *Governance*, supra note 20, at 1097–1105.

lex situs rule, the PRIMA approach, and the Rome and Brussels instruments when confronted with digital assets that lack a stable location, a determinable intermediary, and a legally attributable enforcing agent. The third, constructive claim—that coherent governance requires a transnational conflicts architecture organised around *technological control*, *functional equivalence*, and *layered governance*—has been elaborated through a three-tier conflicts rule applicable to the proprietary, jurisdictional, and insolvency dimensions of decentralised digital collateral.

The insolvency dimension of this analysis deserves particular emphasis as a concluding observation. The Chapter 11 proceedings of Celsius Network LLC before the United States Bankruptcy Court for the Southern District of New York demonstrated, with irreversible consequences for thousands of creditors across jurisdictions, that the absence of coherent PIL rules for digital collateral is not a theoretical lacuna but a practical failure. The question of whether digital assets held in yield-bearing DeFi arrangements constitute property of the insolvency estate or remain the property of depositors cannot be answered by reference to any existing PIL instrument. It can only be answered by the kind of transnational conflicts architecture this article proposes—one that prioritises the concept of *control* over territorial possession, applies functional equivalence to regulatory characterisation, and treats protocol governance as a legally relevant source of normativity rather than a mere technical event.

Decentralised finance does not render law obsolete. The argument advanced here is precisely the opposite: the progressive displacement of institutional intermediaries increases rather than diminishes the importance of legal certainty, because it removes the informal coordination mechanisms—renegotiation, discretionary forbearance, institutional relationships—through which legal uncertainty is ordinarily managed in conventional finance. When enforcement is algorithmic and irreversible, the legal framework governing proprietary rights and jurisdictional allocation must be correspondingly robust. This is not a counsel of technological scepticism; it is a recognition that the rule of law and the rule of code are not alternatives but complements, and that their productive coexistence requires deliberate and sophisticated legal architecture.

The framework proposed here is necessarily provisional. The UNIDROIT PRINS provide a foundation; the European Law Institute's work on digital assets provides an institutional locus; and the growing body of comparative scholarship surveyed in this article provides the doctrinal building blocks. What remains to be constructed is the political and institutional will to translate these materials into binding transnational instruments—whether through a targeted protocol to the Hague Convention, an amendment to the EU Insolvency Regulation, or a dedicated instrument on the conflict of laws applicable to digital asset transactions. The task is urgent. Decentralised lending protocols already manage tens of billions of dollars of digital collateral across jurisdictions. Every day that passes without coherent PIL rules is a day on which legal uncertainty is silently priced into the cost of capital, absorbed by borrowers who do not know what law governs their rights, and deferred to courts and insolvency administrators who will eventually be left to resolve, at great cost, the problems that legal scholarship is capable of anticipating and addressing now.